

B.1 Sampling Universe, Sampling Method, and Expected Response Rate

B.1.1 Sample Universe and Sample Size

The O*NET sampling universe includes total employment in the United States, estimated by the Bureau of Labor Statistics at 140.5 million in 1998, and projected to increase to 160.8 million by 2008¹². O*NET includes 974 occupations, and plans to add information to the database annually based on surveys of persons employed in these occupations, with the goal of updating the information for each occupation every three to five years.

As described in **Section A.2**, O*NET will be used for a wide variety of purposes. The broadest general purpose uses are for career development counseling, workforce development, curriculum development, program administration, and development of labor market and occupational information. More specialized uses relate to activities such as vocational rehabilitation and human resource activities. Given the *O*NET Data Collection Program*'s proposed sample size and occupational level of analysis, these specialized uses may require additional data or validation not available in O*NET before application to job or organization-specific purposes.

The more general purpose uses often involve sorting occupations on specific criteria of interest to a job seeker or individual making a career decision, such as identifying occupations with high ratings on descriptors such as aptitudes, work context and interest scales. This use may also involve comparing the results of an assessment, such as an interest inventory, to the ratings of occupations on the relevant scales. Another approach often used is to provide descriptive information for a specific occupation selected from a list of occupation titles. Some general purpose users may also cluster occupations to broader levels of aggregation than the O*NET occupations.

A key issue in sample design is the level of precision required in the resulting data, and the cost of producing a particular level of precision, both in terms of dollars and respondent burden. The O*NET sample design has been developed to provide results with a level of precision which should be adequate to meet the needs of general purpose users. As described in **Section A.16.1**, release of O*NET data will be accompanied by sampling error measures, allowing users with more specialized purposes to evaluate whether the level of precision is sufficient to meet their needs. Statistical analysis

¹² Occupational Employment Statistics, Bureau of Labor Statistics,
URL:<http://stats.bls.gov/news.release/ecopro.t01.htm>.

indicates that achieving significant increased precision would require a major expansion of the sample size, with accompanying increases in costs and response burden. The additional financial resources are not presently available.

In the Final Technical Report of a previous O*NET project, Peterson, Mumford, Levin, Green, and Waksberg (1997)¹³ state that 15 to 30 incumbents typically provide sufficient inter-rater reliability for describing occupations, given the types of measures O*NET uses to describe people's jobs. This report also presented means and standard deviations for 7-point level scales for the descriptors within *Skills, Knowledge, Generalized Work Activities, Abilities, and Work Styles*. Statistics were computed separately using the reported data for each of six occupations. The standard errors presented in that report were computed for each descriptor within each study domain using the simple random sampling formula, where each of the subjects responded by choosing one of the points of the scale. Means and standard deviations for the 5-point importance scales are also shown in the report for each descriptor in *Skills, Knowledge, Generalized Work Activities, Work Context, Abilities, and Work Styles* and for the same six occupations.

An analysis of the data in these tables indicates that when 15 responses per descriptor are obtained, then the mean values for virtually all of the 5-point and the 7-point descriptors will be estimated within 1 to 1 ½ scale points, with 95 percent confidence, for all occupations. And, as these are maximal values, the mean values of most descriptors would be estimated more precisely (i.e., 68 percent of the descriptors would be estimated within 0.5 to 0.75 scale points). Also, Mumford, Peterson and Childs (1997)¹⁴ cited Fleishman and Mumford (1991)¹⁵ as support that variation of one to one and one-half scale points on a seven-point scale "is typical of that found for well-developed level scales". The instrumentation for incumbent worker respondents consists of four distinct questionnaires. Each questionnaire takes approximately 30 minutes for

¹³ Peterson, N.G., Mumford, M.D., Levin, K.Y., Green, J., and Waksberg, J. (1997). *O*NET Final Technical Report*. Salt Lake City, UT: Utah Department of Employment Security. Chapter 2: Research Method: Development and Testing of the Content Model, pp.19-20.

¹⁴ Mumford, M.D., Peterson, N.G., and Childs, R.A. (1997). *O*NET Final Technical Report*. Salt Lake City, UT: Utah Department of Employment Security. Chapter 3: Basic and Cross-Functional Skills: Evidence for the Reliability and Validity of the Measures, pp. 3-8.

¹⁵ Fleishman, E.A., and Mumford, M.D. (1991). Evaluating classifications of job behavior: A construct validation of the ability requirements scales. *Personnel Psychology*, 44, pp. 523-575.

completion, and each sample employee will only complete one of the questionnaires. The same instrumentation is used for all of the occupations being studied.

The sample size issue was revisited in a statistical analysis conducted recently by Research Triangle Institute (RTI), based upon the data collected in the 1999 O*NET methods pretest. The report on statistical analysis of the pretest data (see **Appendix E**) describes similar results obtained from an analysis of sampling errors computed in accordance with the pretest sample design. The report concludes that samples of 15 responses per descriptor are adequate, and that samples of 10 responses per descriptor even may be adequate, to estimate mean scale values within 1 to 1 ½ scale points.

To insure at least 15 completions per occupation for each of the four job incumbent questionnaires, the target number of completed instruments for the first year of data collection has been established as a total of 21 persons per questionnaire or 84 persons per occupation. The overage is necessary to assure that *at least* 15 completes per questionnaire are obtained. It also allows for the possibility that approximately 5 percent of the completed employee questionnaires may not be useable in the O*NET data set. Thus, plans call for data to be collected from a total sample of approximately 25,200 persons (300 X 4 X 21).

*The O*NET Data Collection Program* will seek information from a sample of employees in approximately 300 occupations each year, and will cover all occupations over three years. The subset of 300 occupations to be surveyed during the first year represents occupations previously identified in high-growth industries, as well as high- and low-technology occupations targeted for early assessment in O*NET. The primary consideration in identifying these occupations for early O*NET data collection was making improvements to existing data that would have the most beneficial impact for customers. Because currency is an issue with all O*NET data, anticipated growth in the workplace was targeted as the impetus for making priority decisions regarding the selection of occupations for study. By targeting occupations with high growth rates, current data could be provided to customers where new job openings are occurring and creating demand for workers with the appropriate skills and knowledges. Other factors considered were the exclusion of data already collected in previous O*NET data collections and the completion of data collection in identifiable sectors already partially covered in the plan (e.g., engineering occupations).

B.1.2 Key Results from O*NET Pretest Survey

The primary purpose of the 1999 O*NET pretest survey was to test new survey and data collection procedures developed to improve the response rates. An experimental design was imposed on the pretest sample, to test the effects of factors thought to influence survey response rates. These experiments were listed in **Section A.1.3**, and details are included in the OMB clearance package for the pretest¹⁶. The detailed statistical results of the pretest conducted by Research Triangle Institute are contained in **Appendix E**. The key findings in this report include the following:

1. The difficulty in locating an establishment that employs the occupation of interest (i.e. eligible for sampling) varied widely across occupational units, but may be predicted based on characteristics of the occupation.
2. Eligibility for sampling can be improved in some cases with a targeted approach and use of multiple sampling approaches in the future may improve data collection efficiency (see section **B.1.3, Use of Multiple Sampling Methods**, which follows).
3. Adequacy of the coverage of an occupation using the targeted approach has not been precisely determined, but the target sampling frames appear to cover roughly 40 to 50 percent of the workforce for most of the pretest occupations.
4. Contrary to expectations, employer response rates were higher from smaller companies.
5. Response rates were better from companies offered the toolkit; this type of incentive is recommended for future use.
6. The number of occupations an establishment was asked about (controlled from one to four) did not appear to be related to the employer response rates.
7. At the employee level, inclusion of the video did not affect the response rate and may be eliminated from future use as an unnecessary expense.
8. A statistically significant effect of the type of return postage was not found. However, the result was borderline and given the potential to increase the employee response rate, we recommend further testing of this methodological factor.

¹⁶U.S. Department of Labor, Employment and Training Administration: O*NET Data Collection Program Survey Pretest, Office of Management and Budget Clearance Package Supporting Statement and Data Collection Instruments, March 1999.

9. Monetary incentives had a significant impact on employee response rates and, as expected, pre-incentives were more successful in encouraging response than were post-incentives.
10. Employee response rates greater than 50 percent were generated using the \$10 pre-incentive; this response rate was indistinguishable from the more expensive plans.
11. Differences were noted between responding and non-responding establishments when examined by industry, geographic region and number of employees.
12. The sample size analysis confirms the adequacy of 15 responses per descriptor, and shows that samples of 10 responses also may be adequate, but samples of 5 responses would be inadequate.
13. The stratification and proportional-to-size sample allocation plan for the complex sampling design employed in the pretest generally appears to have improved the efficiency of data collection compared to a simple random sample.
14. Unbiased estimates of the item means and standard errors obtained with the pretest sample design are now available for researchers and other data users for the 50 occupational units studied in the O*NET pretest.
15. No difference in raw response rates was detected for the type of questionnaire the employee was asked to complete.
16. There was no significant item non-response on the individual questions. Item completion rates ranged from 92 to 100 percent, averaging 99 percent.

In addition to the key statistical findings listed above, a number of other important lessons were learned during the pretest survey. These results and their implications for future O*NET surveys are described below:

1. O*NET should incorporate use of the best design factors from the pretest experiments into future surveys, to maximize response rates and data quality. Future O*NET surveys should use employer incentives, \$10 per employee incentive paid in advance, and no video for employees. Research should be continued to determine which form of return postage provides the better employee response rate.
2. The total number of completed questionnaires per occupation was not controlled well in the pretest. This was primarily due to large variations across occupations in their eligibility rates and response rates. The use of predicted eligibility and response rates to allocate the general sample establishments to industry strata should increase the control over the number of completed questionnaires per occupation.

3. Because very low pretest eligibility rates were encountered in the general sample for some occupations, supplemental employer samples targeted toward specific occupational employment were incorporated into the O*NET pretest. The use of target samples successfully increased the eligibility rates compared with the general sample approach for those occupations. Future O*NET surveys should use target samples to increase efficiency for occupations with predicted low eligibility or response rates. In some cases where it is feasible, O*NET will also use association lists or other employee based lists to take advantage of sampling job incumbents directly in a single-stage data collection process. However, we expect to use this approach for relatively few occupations due to limitations of list frame coverage of total employment in the occupation.
4. The BLS Industry-Occupation Matrix File contains occupational employment information by industry. The industry classifications are quite broad (3-digit SIC or less detail), there are numerous roll-up industry classifications, and employment is suppressed in many of the cells, which contributes to low eligibility rates for some occupations with a general sample approach. O*NET project staff are consulting with BLS staff to obtain more detailed sampling frame information, and to obtain BLS input on sampling occupations predicted to be problematic.
5. A sample based on the BLS Industry-Occupation Matrix File affords adequate coverage of the employment for most occupations, but it is inefficient for sampling some occupations. Other more targeted frames (association, licensure, or commercial lists) are more efficient for sampling single occupations but population coverage is typically poor — 50 percent or less of the employment for many occupations. Therefore, future O*NET surveys should use a dual frame survey design¹⁷ to increase survey efficiency for a few occupations with below average eligibility and response rates.
6. The number of sampled occupations per establishment was experimentally varied from one to four in the pretest, to test its effect on establishment response rates. Analysis indicates that it is feasible to ask establishments about their employees in a larger number of occupations, up to some reasonable limits. O*NET will ask establishments about their employees in up to 10 different occupations in random sequence, until a large enough number of occupations has been identified to allow selection of approximately 15 employees from most establishments sampled. This is expected to increase the eligibility rates for sample establishments, thus increasing survey efficiency without negatively affecting the establishment level response rates.

¹⁷Hartley, H.O. (1974), "Multiple Frame Methodology and Selected Applications". Indian Journal of Statistics, Vol. 36, Series C, Pt. 3, pp. 99-118.

7. The O*NET survey burden on small establishments did not affect the business response rates in the pretest, as the highest rates were obtained from the smaller establishments. The burden of preparing lists of employees in several occupations is obviously less in smaller businesses. A significant proportion of the employment for many occupations is found in small businesses. Thus, O*NET plans to sample all sizes of establishments in future O*NET surveys.
8. Self-employed workers were excluded from the pretest, but should be covered in future O*NET surveys because they account for a sizeable percentage of the employment for some occupations. Efforts will be made to determine the feasibility of sampling self-employed persons and the adequacy of the coverage afforded by the various sampling frames. Self-employed persons who are identified in the O*NET sampling operations will not be excluded from the data collection.
9. There is sufficient empirical evidence to conclude that perhaps as many as one-third of the selected employees in the pretest never received the questionnaire packages. This happens when the point-of-contact (POC) at the sample establishment does not pass the packages on to the sampled job incumbents. This POC nonresponse reduces the employee response rate drastically. Measures are needed to encourage the establishment POCs to distribute O*NET questionnaire packages to their selected employees. Future O*NET surveys will include strong efforts to streamline the POC advance packages and respondent packages, to select employee-based samples when feasible, to provide appropriate employer and POC incentives, and to increase outreach efforts to relevant business and industry groups.

B.1.3 Use of Multiple Sampling Methods

Pretest Sample Designs

Again, the primary purpose of the O*NET pretest survey was to test new survey and data collection procedures designed to improve the response rates. For the pretest survey, both a general sample and several targeted samples were tested.

The ***general*** sample design used in the O*NET pretest was a two-stage design of establishments and their employees. A single sample was used to represent all 50 occupations. Establishments were stratified by industry and number of employees and a composite size measure was used to allocate the sample to strata. The sample establishments were selected with probabilities proportional to total employment and employees were sampled in from one to four occupations per establishment.

The **target** sample design used in the pretest also was a two-stage design of establishments and their employees. During the early pretest data collection, the general sampling approach identified too few establishments with employees in some of the 50 occupations. These occupations were reviewed to determine if finer industry detail than 3-digit SIC was appropriate, and additional establishments were sampled within more detailed industry groups which were expected to be “richer” in the occupations of interest. These establishment samples were thus “targeted” towards the occupations being surveyed. In most cases a separate sample of establishments was used to represent a single occupation. Establishments were stratified by SIC and number of employees; sample establishments were selected with equal probabilities within SIC strata. It should be noted that while the pretest target samples were all two-stage samples of establishments and their employees, we also plan to use single-stage target samples based on association or other list frames when available. This will be done for selected occupations where the list frames provide adequate coverage of occupational incumbents.

Multiple Methods Research

The O*NET project team researched various sampling methods and strategies during the past year. The goal was to identify a combination of methods that would improve sampling efficiency, improve response rates and minimize the burden of future O*NET surveys on the public. Alternative sampling frames were examined, including occupation-based lists of employees available from professional associations, licensing authorities, and commercial list vendors. Higher response rates are expected when it is possible to sample employees in an occupation directly from a list, as compared to two-stage sampling of employers and their employees. Employee-based lists are not available for many occupations, however, and many of the available lists are deficient in their coverage of the occupational employment, their recency, and their specificity.

B.1.4 The O*NET Data Collection Program Sample Design

A multiple frame sample design will provide the general model for the O*NET sample design. This model includes four types of sampling frames:

- # General Employer Sample: the primary method, using a two-stage sample of employers and employees, and where employers are sampled for numerous occupations. This frame will be used for most occupations.

- # Targeted Employer Sample: a two-stage sample of employers and employees, where employers are sampled for one or two occupations but at a more detailed SIC level.
- # Association Member Sample: an alternative to the General Sample, where only association members are surveyed.
- # Dual-Frame Sample: a combination sampling approach using two or more frames, most often either the Association Member Sample or the Targeted Employer Sample in combination with the General Sample.

In addition, Subject Matter Experts (SMEs) will be used to obtain the data for a very small number of occupations in which all four of the sample choices are problematic — for example, occupations with very small employment, new and emerging occupations that do not yet have industry employment data, and those whose incumbents are in remote locations that are difficult to access. To decide which sampling method to use for each occupation, O*NET staff will compare the advantages and disadvantages of each potential method. For each of the occupations, information on the predicted eligibility rate and the predicted response rates will be used to quantify the efficiency of sampling the occupation through a general employer sample. The projected coverage afforded by association lists of employees, when available, will be used to quantify the potential benefit of sampling from such lists. Since a sample represents only the portion of the universe covered by the sampling frame that was used, it is usually not worthwhile to use frames that cover a small part of the population of interest. For this reason, lists covering less than 40 to 50 percent of the employment for an occupation will not normally be sampled in O*NET surveys.

Occupations will be assigned to sampling methods as summarized by the decision chart in **Exhibit B-1**, to take advantage of the strengths of the alternative methods. That is, for occupations with association member lists that cover a high percentage of the occupational employment, O*NET will use the association member sample alone to take full advantage of higher response rates obtained with single-stage designs. For occupations where the association list frame coverage is low (or no list frame is available) and the eligibility rate predicted for the general employer sample is above average, O*NET will use only a general employer sample to take advantage of its overall efficiency. Otherwise, O*NET will use a dual frame sample design involving a general employer sample in combination with either an association member sample

Exhibit B-1. Decision Chart, Sample Design for an Occupation

Association List Frame Coverage	Predictions for General Employer Sample			
	Low Eligibility, Low Response Rate	Low Eligibility, High Response Rate	High Eligibility, Low Response Rate	High Eligibility, High Response Rate
Not Available	Dual Frame Sample	Dual Frame Sample	General Employer Sample	General Employer Sample
Low (below 40 percent)	Dual Frame Sample	Dual Frame Sample	General Employer Sample	General Employer Sample
Medium (40 to 79 percent)	Dual Frame Sample	Dual Frame Sample	Dual Frame Sample	Dual Frame Sample
High (80 percent and higher)	Association Member Sample	Association Member Sample	Association Member Sample	Association Member Sample

or a targeted employer sample. These choices take advantage of higher response rates from the association member samples, higher eligibility rates from the targeted employer samples, and the higher population coverage from the general employer sample.

The sample design and selection process that will be utilized for each sampling method have been constructed to take full advantage of the frame information available for each methodology. These designs will be discussed in detail in the next sections. In summary,

General Employer Sample Design

► Stage 1: Selecting Establishments

Phase 1: Selecting Establishment From List of Approximately 10 Million

Frame: InfoUSA List of all U.S. business locations
 Stratification: SIC and establishment size
 Selection: A large sample of ~450,000 establishments will be selected using stratified, simple random sampling. This selection is done to minimize costs associated with establishment frame construction.

Phase 2: Selecting Establishments From Phase 1 Sample

Frame: Phase 1 sample. Auxiliary information on establishments merged on for use in subsequent selection and stratification.

Stratification: 3 digit SIC or groups of 3 digit SICs; also possibly by establishment size (i.e. number of employees) and geographic region.

Selection: Establishments selected with probability proportionate to size and minimum replacement. Establishment size is total employment or square root of employment. Total number of establishments selected proportional to composite size measure formed at strata level.

► **Stage 2: Selecting Employees**

Frame: Establishments selected from Phase 2

Stratification: From 1-5 occupations selected for each establishment

Selection: Up to a total of 15 employees randomly selected from each establishment

Targeted Employer Sample Design

Analogous to General Employer Sample Design except Stage 1, Phase 2 stratification will be 4, 5 or 6 digit SIC. Due to the “targeted” nature of the design, at the second stage of selection generally no more than 2 occupations will be selected for each establishment.

Association Member Sample Design

Frame: Membership list obtained from associations.

Stratification: Depends on membership information available from association.

Selection: Employees selected from membership lists using single stage, stratified simple random sampling.

Dual Frame Sample Design

Combination of General/Targeted Employer Sample and Association Member Sample.

In summary, the O*NET sample design that will be used for each occupation will be tailored to the coverage that can be obtained from available list frames, taking into consideration the predicted eligibility and response rates for the occupation in a general sample. Many of the occupations will be sampled using only the general sample, the most efficient method for occupations with above average general sample eligibility rates. A few occupations will be sampled using only an association member sample approach, to take advantage of the higher response rates obtained with single-stage sampling of employees. A moderate number of occupations will be sampled with a dual frame design, taking advantage of higher eligibility rates from targeted employer samples and higher response rates from

association member samples, while using the general employer sample to cover the incompleteness of the other frames.

B.1.5 Design of the General Employer Sample

The O*NET general employer sample will be a national probability sample of business establishments and their employees. The first stage sample of business establishments will be selected in two phases; the first will be a large stratified sample from a national business establishment frame, followed by a second phase stratified sampling from the first phase sample that will be augmented with occupational information and design measures of size from other sources. A first phase sample of establishments will be selected from the InfoUSA list of all U.S. business locations. This first phase selection will be a very large sample of establishments selected just prior to beginning the first year's data collection. The second phase sample establishments will be selected with overall probabilities proportional to number of employees, or possibly proportional to the square root of the establishment's employees. The study design is described here as a stratified two-stage probability sampling design in which the first-stage units are business establishments and the second stage units are employees of those establishments.

Data from the most currently available Occupational Employment Statistics (OES) data file will be used to determine which industries to sample for each occupation and the SIC stratification, and to allocate the sample to the SIC strata based on a composite size measure similar to that described in Folsom, Potter, and Williams (1987)¹⁸. The composite size measure is used to achieve a desired sample size allocation for each occupation simultaneously.

The allocated number of business locations then will be selected using a probability minimum replacement procedure developed by Chromy (1979)¹⁹, with selection probabilities within each stratum proportional to the number of employees at the location. By allowing multiple selections of a unit, Chromy's minimum replacement procedure guarantees that the difference between the expected and realized selection frequencies is less than unity regardless of the sample allocation and the sizes of the

¹⁸ Folsom, R.E., F.J. Potter, and S.R. Williams, 1987, *Notes on a Composite Size Measure for Self-weighting Samples in Multiple Domains*, Proceedings of the Section on Survey Research Methods, American Statistical Association, pp. 792-796).

¹⁹ Chromy, J.R. (1979). *Sequential Sample Selection Methods*, Proceedings of the Section on Survey Research Methods, American Statistical Association, pp. 401-406.

units. This is a desirable property for business surveys that employ size measures based on numbers of employees.

First Phase Sample of Establishments

A very large first phase sample will be purchased in the Winter of 2001, for use in selecting establishment samples for the first year's data collection. Commercial list frames have been evaluated, and the InfoUSA frame meets the study requirements; that is, it has the essential establishment-level information including industry type and location-specific employment. The frame contains approximately 10 million establishments and is updated regularly.

Constructing the first stage of the O*NET general sample will involve a number of steps. The first phase sample will be a very large stratified sample of business establishments from the InfoUSA total frame. The cost of purchasing the entire InfoUSA frame would be prohibitive, which provides the motivation for the two phase sample design. To produce this first phase sample, InfoUSA will select approximately 450,000 establishments from their total frame of business locations. Since the ultimate sample will be allocated to establishment strata in proportion to employment, at this point larger establishments will be sampled much more heavily than smaller establishments. This first phase sample of business establishments will be selected systematically and with equal conditional probabilities within size strata. An implicit geographic stratification of the establishments will be imposed by ordering the establishments by zip codes prior to the systematic selection. The sampling intervals will vary from approximately one in 200 for the smallest size stratum to one in one for the larger size strata.

Determining the Industries to Sample

To select industries to sample, we will first identify the OES code (or codes) that relate to each occupation. Then, for each OES code, we will select the smallest set of SIC codes that gives at least 80 percent coverage of the employment for that OES code. This provides a set of SIC codes that will be sampled for each occupation. Each SIC code may be related to multiple OES codes, and most SIC strata will consist of all OES codes that are related to a single SIC. However, some SICs in which the number of employees to be surveyed is very small will be grouped with similar SICs to form an SIC stratum.

First, we will determine which types of business locations, classified by three-digit Standard Industrial Classification (SIC) codes, contain significant concentrations of the occupations.²⁰ The InfoUSA establishment list contains total employment and six-digit SIC codes at the establishment level, but no information about employment by occupation. For this reason, 1998 employment data by SIC and occupation from the Bureau of Labor Statistics (BLS) national OES data file will be used to determine which SICs to sample, and to allocate the sample to SIC strata. Each of the occupations to be surveyed with the general sample will be related to OES occupation codes. (Each OES code covers one or more O*NET occupations. The OES file contains occupational employment information by SIC and OES codes, providing the link between occupations and industries.)

The employment data for all SICs with the relevant OES codes will be extracted from the OES file and examined to determine which SICs will be sampled to represent each of the OES codes (and, in some instances, the sampled O*NET occupations within the OES code). We will rank the SICs within each OES by the employment information to determine how many SICs would be sampled to cover at least 80 percent of the employment of each OES while minimizing the number of SICs sampled. It is important to cover a high percentage of OES employment in order to avoid non-coverage biases in the resulting study data. We will tabulate the percentage of the total employment of each OES that would be represented by the SICs that each contain varying proportions of the OES employment. We will choose a cutoff value for each OES, to minimize the number of SICs sampled for the OES while including sufficient SICs to cover at least 80 percent of the OES employment.

First Stage Sample Allocation

In this section we discuss calculating the composite size measure values, forming the SIC strata, and allocating the establishment samples to SIC strata. Composite size measures will be computed for each OES with sampled employment in each of the associated SICs, based on the 1998 OES employment information. The composite size measure determines the sampling fractions needed to obtain the desired number of completed questionnaires from each of the occupations sampled, over all SICs. Note that the general employer sample will not be used to survey a few occupations, as described in the **Section B.1.3**, “Use of Multiple Sampling Methods.”

²⁰More detailed SICs will be used when the detailed OES data are obtained from BLS.

First, the desired number of general sample completed questionnaires C_j will be determined for each OES related to one of the selected O*NET occupations. The value for an OES depends on the number of occupations being sampled within the OES and is equal to the total number of completes desired from the general sample for those occupations. In most cases C_j will be equal to 84 times the number of sample occupations related to the OES. Note that some O*NET occupations are nested within OESs, so that occupations are encountered at their population prevalence (i.e., welder within metal fabricators). Then a size measure S_{θ} will be computed for each OES ($j= 1,2,\dots$) and for each selected SIC ($i= 1,2,\dots$) as

$$S_{\theta} = \frac{C_j P_{\theta}}{K_j LI_j RI_j L2_j R2_j} \quad \text{where}$$

C_j = the number of completed interviews desired from the general sample for occupations related to OES-j

P_{θ} = the percent of the total OES-j employment related to SIC-i

K_j = the percent of the total OES-j employment included in the SICs being sampled for OES-j

LI_j = the predicted establishment eligibility rate for occupations being sampled from OES-j

RI_j = the predicted establishment response rate for occupations being sampled from OES-j

$L2_j$ = the predicted employee eligibility rate for occupations being sampled from OES-j

$R2_j$ = the predicted employee response rate for the occupations being sampled from OES-j.

If the four predicted eligibility and response rates were all equal to one, then summing the S_{ij} values over the sampled SICs would simply yield the total general employer sample allocation to any OES-j ,

$$S_{+j} = \sum_i S_{ij} = \frac{C_j}{K_j} \sum_i P_{ij} = C_j .$$

the desired number of completed questionnaires from OES-j. Using the predicted eligibility rates and response rates in the size measure calculation increases the allocation of sample to OES/SIC cells according to the expected sample loss from those factors. The OES/SIC size measures will be used to group SICs (indexed by i) to form SIC strata (indexed below by h). Most SICs will form an individual stratum, but some SICs with very small sample allocations may be grouped with similar SICs. The allocation of sample establishments to the SIC strata will be computed in proportion to the S_{i+} values where

$$S_{i+} = \sum_j S_{ij} .$$

The first year general sample will involve selecting n_h establishments in stratum-h and m_{hk} of the eligible employees within establishment-hk. Note that the “i” index for SICs has been grouped to form sample industry strata indexed by “h”. The number of establishments actually needed for the study sample in stratum-h will be affected by the factors of sample loss, non-response, and the distribution of employees by occupation across establishments of the stratum. Sample loss includes establishments that are ineligible for the survey due to no longer being in business or having no employees in the selected occupations of interest, and employees who are ineligible for the survey. Non-response, on the other hand, includes eligible establishments and employees who are selected for the study sample but which do not complete the survey materials. Sample loss is not included in response rate calculations. These factors are generally unknown but may be estimated based on previous work and experience.

The actual sample loss, non-response, and average number of employees sampled per establishment will vary by stratum, occupation, and other factors. For this reason it is essential to have

the capability to adjust the number of in-sample establishments during the survey proper. This will be accomplished by selecting a sample of establishments which is much larger than the expected requirement, dividing this large sample into waves, and sequentially fielding only the number of waves needed to meet the actual sample size requirements, stratum by stratum. Each wave singly (or in combination with other activated waves) will be a probability sample of all business locations.

The basic reason for using sample waves is to enhance the ability to control the realized number of observations more closely, since only the assigned waves need be fully worked. Waves not assigned are held as reserve sample, in case some of them are needed to obtain the required number of observations. By monitoring the survey progress in terms of the number of businesses recruited and the number of workers selected by occupation, we will have more control over the number of completed questionnaires for each of the occupations sampled. We will assign or activate sample waves at random, so only the waves which we assign for recruiting business establishments will be used. The assignment of waves may be done independently for each of the SIC strata. All waves that are activated will be fully worked.

First Stage Sample Selection

After the first phase sample is selected, the establishments will be stratified based on the definitions of the SIC strata. This SIC stratification will provide control over the distribution of the final first-stage sample with respect to the occupations sampled. Within SIC strata, business establishments will be selected such that their sample inclusion probabilities are in proportion to the total number of employees at the location (sampling with probabilities proportional to the square root of the location's employees is also being considered to moderate the effects of large changes in the employment at the establishment level).

A probability minimum replacement procedure will be used to select a sample of business establishments from the first phase sample. Each business establishment will be assigned a size measure T_{hk} based on the number of employees at the location and the establishment's phase one selection probability (i.e., the probability of its selection in the first phase sample):

$$T_{hk} = \frac{E_{hk}}{PI_{hk}} \quad = \quad \text{the number of employees at establishment-k in stratum-h divided by the phase one probability of selection for establishment-k in stratum-h.}$$

The expected sample size $E(n_{hk})$ for establishment-k of stratum-h will be assigned as a function of the expected sample size for the stratum and the size measure for the establishment. That is,

$$E(n_{hk}) = E(n_h) \frac{T_{hk}}{T_h} \quad = \text{the expected sample size for stratum-h times the size measure for the establishment, } T_{hk}, \text{ divided by the total of the size measures, } T_h, \text{ for all stratum-h establishments.}$$

In the selection, the phase one establishments will be ordered by their zip codes within strata, imposing an implicit geographic stratification.

Chromy's probability minimum replacement procedure²¹ allows multiple selections for those establishments with values of $E(n_{hk})$ greater than one. If the expected frequency is greater than one, then the number of sets of employees sampled will be equal to the realized selection frequency (an integer above or below the non-integer expected selection frequency). The $E(n_{hk})$ values which are less than or equal to one are also the selection probabilities for those establishments.

Second Stage Allocation And Selection

The second stage sampling units are the individual employees in selected occupations at the sampled business locations. Preparing lists of all their employees in certain occupations is burdensome for the larger establishments since a large number of employees must be listed. Thus, we plan that only three occupation lists will be constructed in the larger establishments, and no more than five occupation lists will be constructed in any sampled establishment. Whenever possible, the goal is to select a total sample of 15 employees representing three or more occupations from sample establishments. Previous O*NET experience has been that business cooperation begins to fall off when more than 15 employees per establishment are sampled. In establishments that have fewer than 15 employees eligible for selection, all eligibles will be selected.

²¹Chromy, J.R. (1979). *Sequential Sample Selection Methods*, Proceedings of the Section on Survey Research Methods, American Statistical Association, pp. 401-406.

Prior to contacting the selected establishments, a maximum of 10 occupations will be selected separately for each sample establishment from the list of all occupations to be sampled in the SIC stratum. The sample occupations assigned each establishment will be selected with probabilities proportional to the within-stratum size measures and randomly ordered after selection. The number of eligible employees in each selected occupation will be determined during the first or second telephone contact with each establishment. During the screening calls, the POCs will be asked about the listed occupations sequentially until at least three, but no more than five, occupations involving a total of 30 or more eligible employees have been identified (a total of 30 or more is used at this point since some of the employees will later be found to be ineligible, reducing our ability to select 15 employees). Once this criterion has been satisfied, the establishment will not be asked about employees in additional occupations.

When contacted, the establishment will be asked to prepare lists of their employees for some of the identified occupations, as follows:

- # if three occupations with eligible employees have been identified and the total number of eligibles is 30 or greater, prepare lists of all eligibles in those three occupations
- # if four occupations with eligible employees have been identified and the total number of eligibles is 30 or greater, prepare lists of all eligibles in those four occupations
- # if five occupations with eligible employees have been identified, prepare lists of all eligibles in those five occupations.

Some establishments will be asked about all 10 occupations, but will have no eligible employees in any of them. Such establishments will be classified as “ineligible.” Some sampled establishments will have employees in only one or two of the 10 occupations; up to 15 employees also may be selected within these establishments.

An information sheet explaining all tasks that would be performed by the POC will be sent to each identified POC well before any recruitment calls are initiated by the assigned Business Liaison (see **Section B.2.1**) . Prior knowledge of the sampling task and procedures will enable the POC to prepare for the telephone discussion with the Business Liaison. The exact allocation of the 15 sample employees to the occupations assigned each establishment will be determined using a computer routine within the survey control system. The Business Liaison will aid the POC in selecting the allocated number of participants from each occupation list, using a random procedure.

Since the first stage sample establishments will be selected with probabilities proportional to their number of employees, fixing the number of second stage units sampled at each location provides an approximately equal probability sample of employees within each occupation. The application of this sample design will result in a national sample of employees, selected with approximately equal probabilities within each of the first year's general sample occupations. Selection of employees with approximately equal overall selection probabilities helps to assure the selection of representative sub-populations of interest.

B.1.6 Design of the Targeted Employer Sample

As discussed in **Section B.1.4**, if the estimated eligibility rate is particularly low for an occupation when sampled from a 3-digit SIC strata and if the occupation is highly concentrated in specific 4, 5 or 6-digit SIC groups then the occupation will be sampled using the targeted approach. This approach is particularly appealing when the occupation is linked to an OES that is concentrated in a 3-digit “roll-up” industry code that might contain a significant amount of employees in occupations not related to the one of interest.

The process of selecting a sample of employees from the more specific, 4, 5 or 6-digit SIC groups will proceed using the same methodology that will be used for the General sample as described above in **Section B.1.5**. Employees will be selected using a two stage sample design where establishments are selected at the first stage and employees at the second.

First Stage Sample Selection

As with the General sample, business establishments in the Targeted sample will be selection proportional to their size T_{hk} where

$$T_{hk} = \frac{E_{hk}}{PI_{hk}},$$

PI_{hk} is the phase one probability of selecting establishment k in SIC strata h from the initial InfoUSA frame and

E_{hk} is the number of employees at establishment k.

Compared to the General sample, in this context strata h will consist of a small number of 4, 5 or 6 digit SIC groups (generally one group) that have been targeted for the occupation of interest.

The expected number of establishments to select from these strata [i.e. $B(n_h)$] will be computed in proportion to the composite size measure for the strata. In other words,

$$B(n_h) = \frac{1}{\bar{m}_h} S_{h+} = \frac{1}{\bar{m}_h} \sum_{\substack{\text{over OES groups } j \\ \text{and SIC' : } i \in \text{SIC strata } h}} S_{ij}$$

Where $\bar{m}_h$ is the average number of employees selected from each establishment in strata h.

Second Stage Sample Selection

Within selected establishments, up to 15 employees in the occupation of interest will be randomly selected. If the targeted SIC contains significant numbers of more than one occupation of interest then the second stage sampling rules as described in **Section B.1.5** will be applied. In other words, an establishment will be asked to roster no more than 3-5 occupations of interest and no more than 15 total employees will be selected from the combined rosters in order to minimize burden for the establishment.

B.1.7 Design of the Association Member Sample

For selected occupations, employees will be selected from associations when they contain a high percentage of the total occupational employment and provided the association is willing to provide a list of its members in usable form to be used as an O*NET sample frame. Professional associations, licensing authorities and commercial companies will be contacted for possible inclusion in the Association Member sample. This sampling methodology is appealing for several reasons:

- # We anticipate higher response rates and higher eligibility rates compared to the General sample.
- # In general, we anticipate most associations will have membership lists available at the person level. Therefore, employees can be sampled using a comparatively simple, single stage design.
- # Employees will be contacted directly, thereby eliminating any gate-keeper effect associated with the establishment contact.

The sample selection procedures will vary between associations, depending on the type of information available on association members. In general, association lists will be sampled using a single stage, stratified simple random sampling approach.

B.1.8 Design of the Dual Frame Sample

For selected occupations, a potentially attractive association might exist but the occupational coverage of the association might be in question. In order to take full advantage of the single stage sample selection techniques associated with the Association Member sample while simultaneously ensuring adequate coverage of the occupation, a Dual Frame sampling approach will be used. For these occupations, a proportion of the sample will be allocated and selected using the Association Member sample methodology as described in **Section B.1.7** and the remaining sample will be allocated and selected using either the General sample methodology or the Targeted Employer sample methodology as described in **Section B.1.5** and **Section B.1.6**. A question asking about membership in the association will be added to the survey instruments for those employees selected from establishments using the General/Targeted Employer sample methodology. Similarly, a question will be added to the survey instrument for those employees selected from associations asking about their current work status (currently working, retired, unemployed, etc.) This information will allow us to compute the joint probability of being selected from both the Association Member sample and the General/Targeted sample. This joint probability will be used to adjust the sample weights for employees who could have been sampled from both frames.

B.1.9 Weighting And Standard Errors

The basic sampling weights will be computed as the inverses of the overall selection frequencies, and the selection probabilities for each selected establishment and each individual participant. The analysis weights for the eligible sample units will be adjusted to compensate for unit non-response, for both establishments and employees. Weight adjustments or imputations to compensate for item non-response within completed instruments are not planned. Weighting class adjustments will be made to compensate for the missing establishments and participants. The first stage adjustments will be defined by industry codes (general sample) and occupations (target samples); the second stage adjustments will be based upon occupations, or groups of occupations. Weights also will be adjusted to correct for duplication between the general and target frames (dual frame samples).

The eligibility status and the response status will be determined for each sample establishment. The companies will be classified as eligible for the survey or not. The sampling weights for the companies found to be not eligible will be set equal to zero. The eligible companies that complete the sampling of employees will be defined as the company-level respondents, and the other eligible companies will be classed as non-respondents. Weighting classes will be defined by industry codes in the general sample and by occupation codes in the target samples. Some industries and some occupations will be grouped together so that each weighting class will involve at least 15 responding sample units.

B.1.10 Expected Response Rates

The *O*NET Data Collection Program* recognizes the importance of obtaining high response rates, to control non-sampling errors and minimize non-response bias. One of the lessons learned from prior O*NET field experiences, noted by Peterson, Mumford, Levin, Green and Waksberg (1997)²², was the difficulty of obtaining high response rates in the O*NET field survey application. The response rates obtained in the earlier O*NET field surveys, especially the employee response rate, were quite low. Even though businesses had agreed to cooperate, it appears that many of them ceased to cooperate when they received the survey materials to distribute to the sampled employees. Completed, usable questionnaires were returned for only 16.1% of the eligible employees sampled in the American Institutes for Research (AIR) baseline survey; a response rate this low has the potential to seriously compromise the parameter estimates.

The eligibility and response rates experienced in the 1999 O*NET methods pretest were also lower than desired. However, one must keep in mind that the primary purpose of the pretest was to determine experimentally those survey design factors that would maximize O*NET response rates. The RTI report *O*NET Pretest Report: Results of Statistical Analysis* (see **Appendix E**) concluded that, had the “best design” been employed in the entire pretest sample, then a 69 percent employer response rate and 55 percent employee response rate would have been obtained. The first column of **Exhibit B-2** shows what the pretest results would have been had the “best design” been used for the entire pretest — that is, assuming that the entire pretest had involved the use of the employer toolkit incentive, the \$10 prepaid employee incentive,

²² Peterson, N.G., Mumford, M.D., Levin, K.Y., Green, J., and Waksberg, J. (1997). *O*NET Final Technical Report*. Salt Lake City, UT: Utah Department of Employment Security. Chapter 2: Research Method: Development and Testing of the Content Model.

Exhibit B-2. Response Rates from O*NET Methods Pretest “Best Design” and Projected for the O*NET Data Collection Program

Survey Phase	O*NET Methods Pretest (“Best Design”)	O*NET Data Collection Program (Projected)
<i>Phase 1. Solicitation of Employer</i>		
Number of Employers Contacted	6,881	12,000
Eligibility Rate	29.9%	66.0%
Number of Employers Eligible	2,055	7,920
Number participating in Employee survey distribution	1,418	6,180
<i>Business Cooperation Rate</i>	69.0%	78.0%
<i>Phase 2. Solicitation of Employee</i>		
Average Employees Selected per Employer	3.28	6.0
Number of Employees Solicited	4,651	37,080
Number of complete employee responses	2,556	25,200
<i>Employee Response Rate</i>	55.0%	68.0%

stamped return envelopes, and not sending the video to sampled employees. Other survey factors, including the employer sample size, eligibility rates and the average number of employees sampled per establishment, were assumed unchanged.

The enhanced survey procedures described in the paragraphs that follow will lead to significantly higher response rates in future O*NET surveys in comparison with the pretest “best design.” The last column of ***Exhibit B-2*** contains the projected response rates for the O*NET data collection program as it moves into operation. The eligibility rate for sampled establishments will be increased significantly by use of more detailed industry/occupation data and by searching for up to 10 occupations in each sampled establishment. The pretest experiment limited the number of occupations sampled per establishment, varying the number from one to four. To the extent that it is possible to use new sampling frames based on association and license lists, direct employee sampling removes this part

of the sample from the establishment eligibility rate computation. Finally, the increased use of dual frame samples in the production study sample designs also will increase overall study eligibility rates. For the reasons enumerated above, an establishment eligibility rate of 66 percent is projected for the operational survey.

Business cooperation rates will be increased due to the longer data collection period, streamlined materials in the packages mailed to sample establishments, enhanced outreach activities and improvement in the project web site. A longer data collection period will allow more efforts to communicate with sample establishments and obtain their cooperation. Anecdotal information from the pretest indicated that some establishments felt that they were receiving too much information, so future mailings will be streamlined. Establishments that are interested in more detail will be referred to the project web site. A business cooperation rate of 78 percent is predicted.

In the pretest, from one to four occupations were assigned to each sample establishment; the number was varied randomly in order to evaluate the effect of the number of occupations on the response rates. Similarly, the total number of employees selected per establishment sampled was varied from 1 to 16 in the pretest. Prior to the pretest, based on experience with other studies, we predicted an average of 5.33 employees would be selected per establishment using these procedures, but 3.28 per establishment was the actual pretest experience. As described earlier in **Section B.1.2**, sample establishments will be queried about their employees in up to 10 occupations in future O*NET surveys, with the goal of selecting the maximum of 15 total employees from as many of the sample establishments as possible.

Simplified “take-all” sampling procedures will be used in the smallest businesses, making participation easier for these businesses. Considering that it will not be possible to select 15 employees in many of the small businesses, we project that an average of six sample employees per cooperating establishment will be selected.

The analysis of pretest data indicates that mailing the O*NET video to selected employees was not effective and may have slightly reduced the employee response rate. Videos will not be included in the employee mailings in future O*NET surveys. The questionnaire cover is being redesigned to be more attractive to employees, which should improve employee response rates. In addition, the amount of detailed information included in the mailing will be reduced, to streamline the mailings to establishments. Increased follow-up activity (more attempts to follow-up with POCs and

the selected employees) spread over a longer data collection period will allow the employee response to build over a longer time period. The use of the prepaid \$10 incentive and stamped questionnaire return envelopes also will help to attain the projected 68 percent employee response rate in the O*NET operational phase.

There is strong empirical evidence that a sizeable proportion of the sampled employers in both the AIR and RTI pretest surveys did not pass the employee packets along to the sampled incumbents. This tendency toward POC non-response on the part of perhaps as many as one-third of the sample establishments drastically depresses the calculated employee response rates. This problem will be reduced through use of the up-front in-kind gift to the POC, as well as enhanced follow-up efforts with the POC. To the extent that association list samples are utilized for selected occupations in future O*NET surveys, eliminating this POC non-response effect will tend to increase the employee level response rates.

B.2 Procedures for the Collection of Information

B.2.1 The Business Recruiting Staff

Highly skilled and highly trained Business Liaisons, some with human resources experience, will be responsible for interacting with businesses during all steps of the project, securing and maintaining their cooperation and participation. The Business Recruitment Manager will make assignments to each Business Liaison by industry group when possible, attempting to match the industry with the background of the Business Liaison. For example, Business Liaisons who have worked at a hospital in the past will be assigned all hospitals that are selected. The rationale for this is that the Business Liaisons will have some prior knowledge of existing protocols within that industry and will be able to communicate and interact more effectively with those staff. This is particularly important since very little of their presentation/negotiation will be scripted. Only one Business Liaison will be assigned to each selected business.

A sophisticated yet user friendly web-based computerized control system will be developed to track all business information gathered by the Business Liaison. This information will be used to prompt future events such as making follow up calls to the POC, generating personalized letters and occupational profiles, and producing project status reports.

B.2.2 The Screening/Verification Telephone Call to Each Selected Business

The Business Liaison will call each selected business that is provided in the sample to determine that each selected business is eligible (i.e., that the business is in operation, it is the same business sampled at that location or a new business doing the same thing at that location, and it does have employees in one or more of the occupations that we are interested in for that business location). The point-of-contact (POC) for this project, the person within the organization who is knowledgeable of the classification of jobs, will be identified during this conversation. The following POC information will be obtained:

- # name and title of the point-of-contact
- # overnight delivery address
- # telephone number
- # e-mail address (if available)
- # fax number

Packaged scripts are not used during any of the telephone calls to businesses. Guidelines are used to outline the necessary information that is to be obtained from each business. Business Liaisons are trained to listen and interact effectively and in their own creative and comfortable style; therefore, reading off a computer screen is not encouraged. All information that is gathered is entered into the control system by each Business Liaison after each conversation with the POC.

B.2.3 Lead Letter to POC

Project information will be sent by overnight delivery to each identified POC within 24 hours of the successful screening call to each business. The means and quickness of the delivery will lend an air of importance and visibility to the project and materials. This also will facilitate the timing of the follow-up phone contacts. An advance lead letter will be sent to the POC prior to the Advance Package. The lead letter will serve to inform the POC of the importance of the study. This letter will be on Department of Labor letterhead.

B.2.4 Advance Package of Project Information to POC

An Advance Package will be sent to the POC three days after the lead letter. The Advance Package will contain more detailed information supplemental to the lead letter.

The following information will be included in the Advance Package:

- # letter from the Department of Labor
- # O*NET brochure
- # project participation information sheet
- # letter from RTI project director
- # Identification Profiles for selected occupations
- # Business Toolkit/incentive
- # POC incentive.

Examples of the information listed above are found in **Appendix C**.

B.2.5 Gaining the Cooperation and Participation of the Business and POC

The Business Liaison will call the POC of each selected business seven days after the Advance Package is shipped. It is extremely important to give the POC adequate time to receive, read, and process the information. During telephone contact with the POC, the Business Liaison will verify the receipt of the Advance Package, verify that he/she is the appropriate person to serve as the POC, verify the existence of selected occupations at the business by discussing in detail the description of each occupation as outlined in the corresponding Identification Profile, explain the O*NET project in greater detail, answer questions, identify barriers to participation and work out a plan to remove any barriers, discuss what is expected of the POC and the selected employees if they choose to participate, and obtain the POC's commitment to participate. All POC refusals will be referred to the Business Recruitment Manager for one refusal conversion attempt (which usually is preceded by multiple calls).

All in all, gaining the participation of the business and POC may take several calls. The Business Liaison will work around the POC's schedule and move at a pace that is comfortable for the POC.

B.2.6 Business and POC Participation in the O*NET Data Collection Program

Once the business expresses interest in participating, the Business Liaison will ask the POC to assist in the sampling of employees in the targeted occupations (that have been verified). It should be noted that *completing the sample* is a key event in that this accomplishment signifies that the business

participated in the O*NET Program. The POC will be aware of this task, as it will have been highlighted in the information sheet sent to the POC. The POC will compile a numbered list of employees' names for each identified occupation (this step may not be necessary in smaller businesses). If three occupations are targeted for a business, then the POC will create three separate numbered lists of employees in those occupations. The POC will tell the Business Liaison how many names are in each list. If there are 15 or fewer employees, all will be surveyed. If there are more than 15 names, this information will be entered into the computer by the Business Liaison and the line numbers of all randomly selected employees in each occupation will be determined. The POC will note the line numbers on his/her lists. The POC will have sole possession of the lists.

If the POC prefers to provide the names of employees and the employees can receive mail at work, RTI will ship overnight the survey packet directly to each selected employee. (This will be based on the POC's assurance that internal policy permits the release of selected employee names and that they may receive mail at the company). The packet will contain a letter from the RTI project director, the assigned questionnaire, a return envelope, an information sheet for completing the questionnaire, and \$10 cash for taking the time to read and process the material. Examples of the letters and the information sheet are provided in **Appendix C**. The questionnaires are provided in **Appendix D**. If RTI does not receive the names, then the individually sealed and labeled survey packet for each sampled employee will be sent overnight to the POC for proper distribution by linking the line number found on each packet label to the occupation lists that the POC maintains. The POC will be asked to communicate to all selected employees, in whatever manner works best, that the company supports O*NET and encourages their voluntary participation.

B.2.7 Employee Participation in the O*NET Data Collection Program

The selected employees will be encouraged to complete the questionnaire on their own time, not company time. The questionnaire will take less than 30 minutes to complete. Each selected employee will return the completed questionnaire directly to RTI. We will ask that they return the instrument to RTI within two weeks.

Sampled employees have the option of completing an online version of the assigned questionnaire by accessing the data collection Web page. This was discussed in detail in **Section A.3**.

B.2.8 Follow-up with the Participating POCs

The Business Liaison will call each of his/her assigned participating POCs seven calendar days after the shipment of the survey packets, regardless of whether they were shipped to the sampled employee or POC. The purpose of this call will be to make sure that the questionnaires arrived and to check the status of distribution activities. If the survey packet was shipped directly to the sampled employee, a letter will be mailed to each sampled employee a couple of days after the shipment informing the sampled employee that he/she should have received the packet by overnight delivery. If the selected employee has not received his/her packet, the letter will instruct them to notify the POC as soon as possible. This will enable the POC to know of any missing questionnaires prior to the telephone call from the Business Liaison. This seven day call also will keep the POC in the loop.

The Business Liaison will call the POC twenty-one calendar days after the questionnaire shipment date to provide a status report on the received and delinquent questionnaires. The Business Liaison will inform the POC that replacement questionnaires will be sent for those that have not arrived at RTI within the following 10 days. Thirty-five days after the initial mailing the POC will be sent a letter indicating which incumbents (identified by their list number) have not returned the O*NET questionnaires yet. Postcards for all of the selected employee incumbents will be included with the letter. The postcards will include both a thank you “if you have already returned the survey” and a request to fill out and return the questionnaire if they have not done so already. The cover letter will ask the POC to distribute the postcards to the selected employees. The POC will be called again 45 days after the initial mailing of packets. The POC will be thanked for their participation and the status of employee participation will be discussed. No more *scheduled* follow up calls will be made to the POC during the data collection period. A letter of appreciation will be sent to each participating business/POC at the end of the data collection period.

B.2.9 Post Collection Data Processing

Approximately 8 weeks will be allowed following the shipment of replacement survey packets for all completed questionnaires to be received at RTI. All questionnaires received by that time will be processed (edited and keyed) and the data file will be developed for later analysis. All questionnaires will be stored in a locked document control area at RTI until federal authorization to destroy them is issued.

B.2.10 Data Collection through Association Lists

For a small number of the occupations, a more direct approach to reach survey respondents will be through professional associations. This may be in addition to the employer-based approach or in place of the employer-based approach, as was discussed in **Section B.1.4**.

Associations will be offered the same incentives as businesses, as will be discussed in **Section B.3.1**.

B.2.11 Subject Matter Experts: Additional Source for O*NET Data Collection

Subject matter experts (SMEs) will serve as the source of data for a small percentage of occupations, including: 1) those with an extremely small population of job incumbents (e.g., with fewer than 10,000 job incumbents, such as Marine Architects [approximately 1,000]); 2) those without industry and employment data (e.g., new and emerging occupations such as Website Developers not currently included within the SOC); and, 3) those where incumbents are determined to be highly inaccessible (e.g., Ordinary Seamen and Marine Oilers). Professional associations, academic/educational organizations, labor unions, licensing bodies, and other professional groups will be used to identify subject matter experts representing selected occupations. A sampling frame of convenience will be developed, from which approximately 28 experts will be selected. Each participating expert will complete the four O*NET survey packets (i.e., Generalized Work Activities, Knowledges, Skills, and Work Context). The goal will be to collect data from at least 21 experts.

Identifying Subject Matter Experts

The professional groups that O*NET identifies as having access to SMEs associated with the occupation (identified for SME data collection) will be asked to develop a list of experts for the occupation. SMEs will be identified based on their experience with the occupation, including serving as a past job incumbent, supervisor, or trainer. Their education related to the particular occupation and their position in a professional association or related group will be considered. While a sample of convenience will be created, emphasis will be placed on developing a sampling frame of SMEs that is representative of key variables that the professional group states may have an affect on the ratings for the occupation (e.g., regional, industry, and/or company size differences may be identified by the professional group). A random sample of 28 experts will be drawn from the sampling frame. The size of this sample assumes an approximately 80 percent response rate from the experts, which will result in an average of 21 complete responses.

This only slightly exceeds the number of questionnaires to be completed by incumbents in the O*NET data collection effort.

Contacting Subject Matter Experts

Sampled SMEs will be contacted by telephone. Contacts will be made by O*NET Business Liaisons. Additionally, in certain cases, a professional association/group may assist with the contact effort. During the initial contact, potential participants will be given an overview of the project and why they are being contacted. An effort will be made to assess their preliminary identification as a subject matter expert (e.g., questions about their education, training and experience). For persons identified as SMEs, their role in the project will be discussed (e.g., serving as a national expert, completing a series of survey instruments). When SMEs agree to participate, the survey schedule will be discussed (e.g., date they will receive packets, expected three week turn-around time). The Business Liaison also will leave contact information in case the experts have any questions. If potential SMEs have further questions about the project, additional project summary information will be sent to them and they will be contacted at a later date to see if they would like to participate.

Administration of Survey Packets

Each SME will receive a set of four O*NET survey packets to complete (i.e., Generalized Work Activities, Knowledges, Skills, and Work Context), along with a background questionnaire and a task questionnaire. The four survey packets will be randomly assigned to respondents to reduce order effects. Additional project summary information, general instructions for completing questionnaires, and mail-back materials will be contained in the SME package. SME packages will be sent out via overnight mail. SMEs will receive a follow-up phone call one week following survey mail-out.

Prior to sample selection, the predicted eligibility model described in **Section B.1.4**, will be used to identify occupations for which the general employer sample method is likely to produce low eligibility rates. These occupations will be examined to determine if there are associations whose membership includes a significant portion of the occupation's job incumbents. For such occupations, either an association list frame or a dual frame sample (involving either a general employer sample or a targeted employer sample) will be used, and the cooperation of the association(s) will be sought for selecting a sample of their members.

During the outreach work, some associations may be identified in which the membership includes a very high portion of the job incumbents for specific occupations. In these cases, the cooperation of the association(s) will be sought for selecting a sample of their members. If cooperation is achieved, data collection for the occupation will be done using only the sample of association members. Once the associations are identified through rigorous guidelines, they will be asked to assist with data collection by making their membership lists available to RTI for distributed surveys.

Extensive research is underway to determine appropriate associations to target for data collection. The focus will be on those associations whose members comprise a large percentage of workers in the occupation, the scope of each of the various associations relating to a particular occupation, the percent of occupational coverage the association membership represents, and the capacity of the association to survey its membership.

The following criteria will be used for evaluating the potential utility of association lists as sampling frames for particular occupations:

- # The professional association's membership represents a large percentage (at least a majority) of the nation's total workforce for that occupation. If the membership does not cover at least 40% or more of the occupational employment, they would have to be excluded from consideration.
- # Adequate information about the members must be made available — at a minimum, name and address. Ideally, information offered by the association would include telephone number, job title, employment status, age, gender, race, and ethnicity, for use with lists that contain retired persons or individuals from more than one O*NET occupation.
- # The professional association is willing to make their membership list available for sampling purposes or will select a sample for the study with our assistance. This factor is critical.

For occupations where association members are surveyed, the questionnaire will be modified to include questions concerning the respondent's employment status. Respondents who are not employed are ineligible for the survey and will be asked to return the questionnaire without completing the occupational questions. Also, where the dual-frame sample is used, employees sampled through contacts with employers will be asked whether they are members of the associations for which members are sampled. This information will be used to compute correct sampling weights, as

employees who are association members potentially could be selected in both sampling frames.

B.3 Methods to Maximize Response Rates

Many enhancements have been incorporated into the *O*NET Data Collection Program* since the pretest to maximize response rates. These include:

- # a non-cash incentive for businesses
- # an in-kind gift to the point-of-contact
- # a cash incentive for respondents
- # minimized burden for businesses and respondents
- # outreach activities including endorsement letters
- # multiple sources for data collection
- # enhanced web-based option for completing questionnaires
- # more attractive appearance of the questionnaire cover.

Many of these enhancements were tested and evaluated for effectiveness during the pretest (see **Appendix E**). Throughout the project, we will undertake efforts to identify and implement additional methods to continue to improve response rates.

B.3.1 Business Incentive

The effectiveness of a non-cash incentive in obtaining the cooperation of businesses in the *O*NET Data Collection Program* was proven in the pretest. Offering the Work Analysis Tool Kit to businesses resulted in a 68% cooperation rate, as compared to 52% cooperation when the tool kit was not offered. We intend to offer a similar non-cash incentive to businesses and associations during the full data collection as a means of maximizing business cooperation rates, as well as an in-kind gift (cash value less than \$10) that will be sent to the organizations' POC along with the advance package of materials.

B.3.2 Respondent Cash Incentive

The pretest evaluated the relevant effectiveness of seven different cash incentive plans on employee response rates (see **Exhibit A-4**). One plan, the \$10 pre-incentive, clearly showed cost effectiveness over the others tested. Per **Section A.9**, we plan to utilize the \$10 pre-incentive for the

full data collection program by enclosing a \$10 bill in each of the initial questionnaire packets (including sampled association members). The POC will not routinely be made aware of the presence of the cash contained in the questionnaire packets. If asked, however, the Business Liaison will inform the POC of the incentive.

B.3.3 Minimized Burden

Anecdotal information received during the pretest indicated that the materials sent to both the business POC and the employee respondents must be streamlined. Therefore, it is important that the *O*NET Data Collection Program* take up as little of their time as possible if we expect businesses and employees to cooperate at the high levels required. One way to do this is to minimize and simplify the informational and instructional materials, as well as the processes, presented to the POCs and employees. For example, there is a “box thud” effect that can discourage a POC from participating. When POCs receive the initial package of materials and thumbs through the folder stuffed with fine print letters, sample questionnaires, ID profiles for multiple occupations, endorsement letters, a brochure, etc., they can perceive the requirements of the study to be too time-consuming from the start. Thus, the materials may be relegated to a low priority pile on the desk and subsequent contact with our Business Liaisons may be avoided.

Building on our pretest experience, we intend to streamline the initial package of materials to POCs. Much of the informational materials that were provided in the POC package during the pretest will be made available on the web for viewing as needed at the POC’s convenience. Further, to avoid overwhelming the POC with a large advance package of materials suddenly appearing without prior notice, we intend to send a short lead letter to introduce the study and advise that a folder of relevant materials will arrive in the next few days. This “spoon-feeding” approach, coupled with the streamlined materials, should effectively minimize the negative impact of the “box thud” effect and, in turn, increase cooperation rates among businesses.

Similarly, we intend to minimize the amount of material and instructions presented to the sampled employees. During the pretest, for example, we evaluated the effectiveness of including a copy of the brief O*NET video tape with the employee’s questionnaire packet. We found no statistically significant difference in the response rates between those receiving the video tape and those who did not. In fact, what small effect was observed was in the negative direction for those receiving the tape. Therefore, a video tape will not be provided to employees during the full data collection.

Lastly, in an effort to further simplify the processes and to decrease the amount of printed instructional documentation needed, we will reduce the number of instances when the employee sampling process must occur. The rostering and sampling process is one of the most burdensome of the POC's activities in the medium-to-large size businesses. For the full study, we are proposing to take a 100% sample of employees in an occupation up to a limit (see **Section B.1**). While the second stage sampling process still will be utilized in larger businesses, the 100% sample option will greatly reduce the burden for medium-to-smaller businesses. The resulting effort should facilitate an increased rate of cooperation with the data collection process.

B.3.4 Outreach Activities

One of the most notable endeavors to assist in achieving the objective of a high response rate is the outreach efforts undertaken by the National O*NET Consortium to target audiences — the business community, the human resource community, the education community, and national stakeholder organizations.

Outreach efforts of the National O*NET Consortium will increase awareness and solicit endorsement of data collection from professional and industry associations. The targeted associations relate to the occupations being surveyed.

During the pretest data collection, the National O*NET Consortium conducted an outreach campaign targeting over 150 associations representing the first 50 pre-selected occupations to be tested. The campaign was a great success, returning 40 endorsements from key associations, including the "umbrella" organizations of the American Business Conference, National Alliance of Business, National Retail Federation, National Coalition for Advanced Manufacturing, National Institute for Metalworking Skills, National Association of Manufacturers, and the U.S. Chamber of Commerce.

In the next phase of data collection, the National O*NET Consortium will continue the outreach campaign aimed at further increasing awareness of O*NET in the business and employer community. Along with awareness-building, the Consortium will strive to demonstrate the importance and possibilities of O*NET, while securing meaningful endorsements from key industry and association leaders. These efforts will enhance the success of all phases of data collection.

As part of the effort to maximize response rates, the National O*NET Consortium developed extensive outreach materials targeting stakeholder communities. These materials will be included in the package sent to the POCs. The new branding package for O*NET created a new look and feel, including a new design and image, logo, and identity. In addition, the National O*NET Consortium built upon this new branding package by developing templates for various target audiences, including internal and external audiences in both the public and private sectors, primarily focusing on the business community, industry associations, human resources leaders, and the education community — those communities that will be contacted for the data collection. Materials produced include:

- # templates to customize existing materials
- # one-page executive brief handouts:
 - *O*NET for Job Seekers: Find the Job That Fits You,*
 - *O*NET for HR Professional: Hire Intelligence,*
 - *O*NET for Business: Helping Businesses Succeed at the Business of Employment*
 - *O*NET In Action: Who’s Using O*NET—And How,*
 - *O*NET’s Impact: What People Are Saying, and*
 - *How Can I Get It?: Where to Find O*NET.*
- # brochures
- # information folders.

In addition, further outreach and media pieces were created to engage professional associations to educate their members about data collection. These materials include:

- # articles for association newsletters and journals
- # bylines for business publications
- # drop-in stories for all media sources, and
- # articles and announcements tailored for association and industry web sites.

Draft letters of endorsement, similar to those used in the pretest, have been prepared that agree to endorse the *O*NET Data Collection Program* and to promote O*NET. The endorsements will be collected and listed on a designed template to be used by RTI in the related survey packets being sent to employers and job incumbents.

For the next phase of data collection, the National O*NET Consortium will continue its awareness building efforts, targeted to the 300 occupations to be surveyed in 2001. These outreach efforts will be based on the lessons learned from the pretest outreach, thereby enhancing our success in raising awareness and garnering support of O*NET. For instance, in the initial outreach campaign, there was an overwhelming call for more information about O*NET before the surveys were sent out. Therefore, by targeting the associations, stakeholders and opinion leaders are reached through association meetings, newsletters, and journals.

For the occupations to be surveyed, the National O*NET Consortium will develop a database of specific contacts and targeted associations at the national, state, and local levels. A campaign strategy using this database will be devised to guarantee coverage of the occupations, including contact with key representatives, members of boards of directors, and committees of associations.

Telephone and in-person contact will be made with appropriate industry or employee related associations to introduce or re-introduce O*NET, to explain the data collection process, and to receive the support of the association. We will request that each of the key professional associations distribute information about O*NET through their member paths, host presentations, place articles in their association journals, and encourage members to participate in the data collection. These associations will be asked to give a meaningful endorsement of O*NET and also to play a role in data collection, if appropriate (refer to **Section B.2.11**). Personalized discussions appropriate for each association, industry or occupation will be conducted.

B.3.5 Multiple Sources of Data Collection

As described in **Section B.1**, multiple sampling methods will be used, which will increase the overall response rates and therefore reduce burden. Also, for a small number of occupations, the Subject Matter Expert method will be used.

B.3.6 Web-based Questionnaires

As a further means of maximizing response rates, we intend to offer a web version of the questionnaires as an option for respondents. Some respondents may find completing the questionnaire electronically via the Internet to be a preferable option to filling out and mailing the hardcopy booklet. Our integrated control system will insure that only one form of a questionnaire is processed from a

respondent. The web data collection process offers several operational efficiencies as well. Therefore, we intend to promote and encourage the use of the on-line questionnaire option by means of highlighting its availability and advantages in the package of materials.

B.3.7 Additional Measures

The following are additional measures that will be incorporated in the *O*NET Data Collection Program* to maximize response rates:

- # Only **one** highly trained and highly skilled O*NET research professional, who is a Business Liaison, will be assigned to interact with each selected business. Many of the Business Liaisons will have human resources experience and may also have specific industry experience. This one-to-one arrangement will build trust, rapport and professionalism.
- # The assigned Business Liaison will not need a script, but rather guidelines to gain the cooperation and participation of the POC/business. Scripting may interfere with the interaction process.
- # The POC will receive a personalized letter from the RTI project director, as well as letters of support for O*NET from agencies and professional associations, and an information sheet explaining exactly what POC and employee participation entails. Additional information will be available for viewing or downloading as needed on the project web site. The idea here is to equip the POC with information and knowledge regarding O*NET and what we are asking them to do. All information will be sent overnight to the POC.
- # Non-cash incentive(s) for participating businesses will be available (see **Section A.9**).
- # An in-kind gift will be given to the POC.
- # The employee sampling procedure will be very simple and the employee selections will be made by computer, based on input from the POC and with the assistance of the Business Liaison.
- # There will be as much interaction as needed between the Business Liaison and the POC. We are dedicated to taking a very hands-on approach and work with each business at their schedule and speed.
- # The POC may not have to distribute the questionnaires to employees. If it is permissible to release the names of selected employees, then the instruments will be

mailed directly to the employee in a packet also containing a personalized letter from the RTI project director, an O*NET brochure, applicable endorsement letters, instructions for completing the questionnaire, monetary incentive for taking the time to review our study materials, and a postage-paid return envelope. (Note: During the pretest, however, only 39 of the 3,766 questionnaire packets were shipped directly to employees.)

- # As an option, an online version of the instrument will be available for completion via the Internet.
- # We will **not ask** the POC to be responsible for collecting completed questionnaires. Employees will be asked to mail them back to RTI in a postage-paid return envelope that is provided. This procedure will further enhance our assurance of confidentiality and will most likely help to increase the response rate.
- # We are encouraging the employees to complete the questionnaire on their own time, so as not to interfere with their work.
- # We will follow up with the POC a couple of times by phone for nonrespondents. We will resend the outstanding questionnaires after 31 days. Subsequently, we will send the POC a set of thank you and reminder postcards to distribute to the selected employees. We do not want to pressure or harass the POC, for that approach likely will **not** yield better results.

Non-participating establishments will be tracked at the business level. The Business Recruitment Manager will attempt all POC refusal conversions. The reasons for refusals will be well documented in the control system and systematically reviewed. This systematic examination will allow program staff to make improvements to the recruiting procedures where possible.

For participating businesses, a follow-up call will be made to the POC thirty-one days after the initial mailing of the employee survey packets (see **Section B.2.8**). Business information that is included in the sample file will be analyzed for possible patterns in the nonrespondent population.

B.4 Tests of Procedures

During the pretest, one of the methodological experiments dealt with the effect on response rates of using first class stamps versus business reply on the return envelopes.

The literature²³ suggests that an increase of two-to-four percentage points in response rates can be achieved by using the first class stamp approach. There are cost factors associated with the use of first class stamps, however, not the least of which is that the return postage cost is incurred for 100% of the sample in the outgoing mailing. Using the business reply envelopes, the postage cost is incurred only for the actual responses. There is also the cost of manual labor associated with affixing stamps to the envelopes that is not a factor with pre-printed business reply envelopes.

Given the cost factor, we attempted to test the effectiveness of the two types of return envelopes during the pretest. Our results (see **Appendix E**) were statistically inconclusive ($p=0.4$); however, the small difference that was observed was in the positive direction for the first class stamp option as the literature would have us expect. Since the *O*NET Data Collection Program* is designed to be a long term, recurring process, involving significantly large samples, we feel it would be prudent to continue our testing of the return envelope options during the first year of data collection.

As we did during the pretest, we propose to randomly assign one of the two return envelope formats to the sample of employees. Near the end of the first year of data collection, we will analyze the response rates by type of return envelope to determine if a statistically significant difference in the rates can be detected. If so, we will propose using the more effective envelope type for all future data collection beginning in year two.

Other than the type of return envelope, no further tests of procedures have been identified at this time for the *O*NET Data Collection Program*.

B.5 Statistical Consultants

The individuals shown in **Exhibit B-3** have consulted on the statistical aspects of the *O*NET Data Collection Program* sample design and/or will be involved in the data collection and analysis. Those individuals shown in sections 1 and 2 of the table reviewed a draft of the OMB Clearance Package in the summer of 2000. At that time, the package contained a sample design and survey process very similar to that contained in the current request. We have revised

²³Dillman, D.A. 2000. *Mail and Internet Surveys: The Tailored Design Method*. New York: John Wiley and Sons.
Dillman, D.A. 1978. *Mail and Telephone Surveys: The Total Design Method*. New York: John Wiley and Sons.

Exhibit B-3. Statistical Consultants

Name	Organization	Telephone Number
(1) Non-Federal Statisticians and Researchers		
Steve Heeringa	University of Michigan	734-764-4424
(2) Federal Government		
Shail Butani	Bureau of Labor Statistics	202-691-6347
Patrick Cantwell	Bureau of the Census	301-457-8105
Nancy Kirkendall	Energy Information Administration	202-426-1088
Alan Dorfman	Bureau of Labor Statistics	202-697-7378
(3) Data Collection / Analysis Contractors		
Don Camburn	Research Triangle Institute	919-541-6696
John Campbell	University of Minnesota	612-625-9351
Randall Keesling	Research Triangle Institute	919-541-6665
San Keller	Research Triangle Institute	919-541-7051
Paul Moore	Research Triangle Institute	919-485-7720

the package based on their comments. No additional external review of the present package has been conducted by these individuals.

The DOL/ETA officials responsible for the *O*NET Data Collection Program* are Jim Woods (202-693-3641) and Pam Frugoli (202-693-3643).